

Reader Disagreement Identifies Residual Diagnostic Risk in Automated ECG Interpretation

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Abstract

Reader disagreement may preserve diagnostic difficulty lost after adjudication to a single reference label. We evaluated uncertainty-based referral and reader disagreement in CODE-test (827 12-lead ECGs). Overall diagnostic error decreased from 7.4% at full coverage to 3.9% at 90% coverage, but disagreement-associated excess error persisted. At full coverage, error occurred in 33.3% of disagreement ECGs versus 6.3% with consensus (RR, 5.29; 95% CI, 2.70–8.85). From 90% to 100% coverage, RR and PR remained above 1 with 95% CIs excluding 1, indicating residual diagnostic difficulty despite confidence-based referral.

Keywords: electrocardiography, automated diagnosis, reader disagreement, uncertainty estimation, arrhythmia

1. Introduction

Reader disagreement during electrocardiographic (ECG) interpretation may signal cases near diagnostic boundaries, yet conventional evaluation typically reduces the adjudication process to a single final reference label, discarding information about how readily consensus was reached. Automated ECG interpretation has shown strong performance in rhythm and conduction diagnosis, particularly for arrhythmias, and has increasingly been extended to derive clinically relevant information beyond categorical classification [1, 2]. Prior work has demonstrated limited expert agreement in difficult AI–reader discordant ECGs, while uncertainty estimation can support referral of lower-confidence automated interpretations for expert review [3, 4]. We therefore evaluated whether uncertainty-based referral reduced overall automated diagnostic error and whether excess patient-level error risk associated with reader disagreement declined in parallel or persisted across retained coverage.

2. Materials and methods

This retrospective secondary analysis used the Clinical Outcomes in Digital Electrocardiology (CODE) datasets from the Telehealth Network of Minas Gerais, Brazil, comprising standard 12-lead ECGs acquired predominantly with TEB ECGPC or ErgoPC 13 devices [1]. CODE-15%, containing 345,779 ECGs from 233,770 patients recorded between 2010 and 2016, served as the model-development cohort [5]. Independent evaluation used CODE-test, an arrhythmia- and conduction-focused set of 827 ECGs from 827 patients collected in 2018. Each ECG was independently interpreted by 2 cardiologists for 6 pre-specified rhythm and conduction diagnoses, with disagreements adjudicated by a third senior specialist [1].

An ensemble of 5 lightweight 1-dimensional convolutional neural networks with the same architecture was trained using different random seeds, and predicted probabilities were averaged. Following prior work, diagnosis-specific confidence was defined relative to the decision threshold, and each ECG was assigned the confidence of its least confident diagnosis [3]. Progressively lower-confidence ECGs were referred from automated interpretation.

Error burden was the proportion of retained ECGs with at least 1 incorrect diagnosis among the 6 labels. At each coverage level, RR compared error risk between ECGs with reader disagreement and consensus, while PR quantified the enrichment of reader disagreement among error-containing ECGs relative to all retained ECGs. Confidence intervals were obtained from 1000 whole-ECG bootstrap resamples using the percentile method, with ratios recalculated within each replicate.

3. Results

Diagnostic errors were concentrated in the lowest-confidence ECGs. At full coverage, 61 of 827 ECGs (7.4%) contained at least 1 diagnostic error, decreasing to 40 ECGs (5.1%) at 95% coverage and 29 ECGs (3.9%) at 90% coverage (Fig. 1). Thus, referral of lower-confidence ECGs progressively reduced the retained error burden.

Across 80% to 100% coverage, ECGs with reader disagreement maintained a consistently higher patient-level diagnostic error rate than those with reader consensus (Fig. 2A), rather than the excess risk emerging only as lower-confidence ECGs were added.

Given that only 33 ECGs had reader disagreement, ratio estimates became increasingly imprecise at lower retained coverage, as reflected by the widening confidence intervals. From 90% through 100% coverage, however, both the RR and PR remained above the null value of 1, with their 95% confidence

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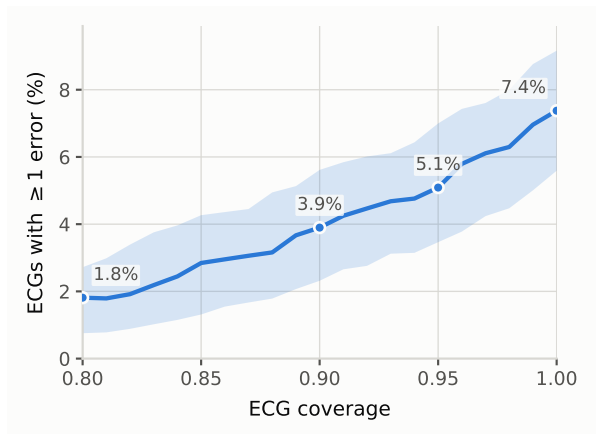


Figure 1: **Uncertainty-based referral of whole ECGs.** Error burden across uncertainty-based whole-ECG referral, using the least confident diagnosis to define ECG-level confidence. Annotated points indicate selected coverage levels and shading represents 95% ECG-level bootstrap confidence intervals.

intervals also excluding 1 (Fig. 2B). Thus, reader disagreement remained associated with greater diagnostic error risk and was consistently enriched among error-containing ECGs even after lower-confidence ECGs were referred.

At full coverage, any diagnostic error occurred in 11 of 33 ECGs with reader disagreement (33.3%) compared with 50 of 794 ECGs with reader consensus (6.3%; RR, 5.29; 95% CI, 2.70–8.85; two-sided Fisher exact $P < 0.001$). Reader disagreement occurred in 33 of 827 ECGs (4.0%) overall but in 11 of 61 ECGs containing any diagnostic error (18.0%; PR, 4.5; 95% CI, 2.6–6.8). In 9 of the 11 ECGs with both reader disagreement and diagnostic error, the error involved the disputed diagnosis; in the remaining 2, it involved a different diagnosis.

4. Discussion

Uncertainty-based referral reduced overall error burden but did not eliminate the excess risk associated with reader disagreement. Disagreement therefore marked a subgroup with residual diagnostic difficulty even after lower-confidence ECGs were excluded. It should not be interpreted as evidence that the adjudicated reference was incorrect, but rather as a marker of an interpretive boundary where independent cardiologists did not initially converge, consistent with prior expert readjudication of difficult AI–reader discordant ECGs [4].

These findings distinguish two complementary signals in automated ECG evaluation: model confidence can identify interpretations suitable for referral, whereas reader disagreement preserves information about reference-label reliability that is lost when adjudication is collapsed to a single final label. Preserving individual reader labels and adjudication history may therefore improve evaluation of automated ECG systems. This retrospective analysis included a modest number of disagreement ECGs, and the findings warrant confirmation in larger cohorts.

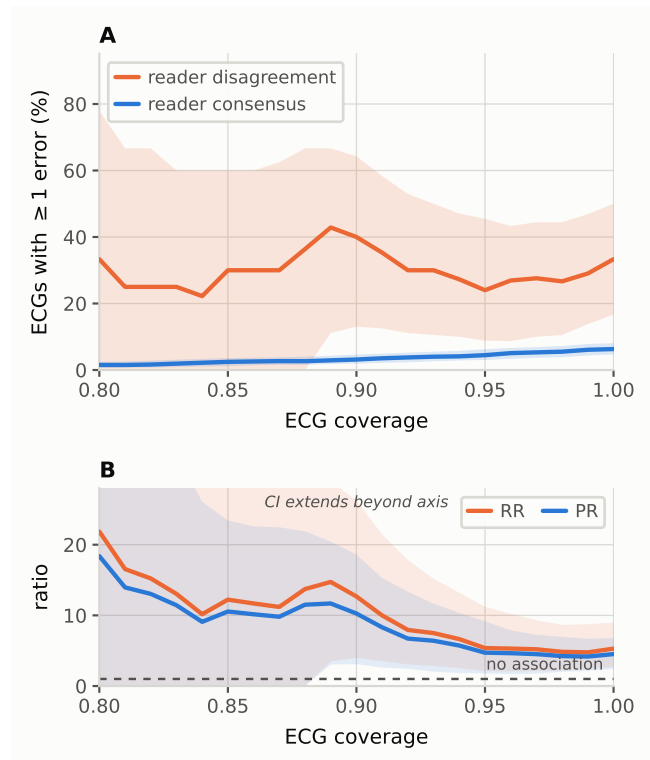


Figure 2: **Reader disagreement and residual diagnostic risk across uncertainty-based referral.** (A) Patient-level diagnostic error across retained ECG coverage, stratified by reader disagreement versus consensus. (B) Risk ratio (RR) for any diagnostic error and prevalence ratio (PR) for enrichment of reader disagreement among error-containing ECGs, evaluated across retained coverage. Shading represents 95% bootstrap confidence intervals; the dashed line in panel B denotes the null value of 1.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRedit authorship contribution statement

Khanh Duy Tran: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no competing interests.

Data availability

CODE-15% and CODE-test are publicly available from Zenodo (doi:10.5281/zenodo.4916206 and doi:10.5281/zenodo.3765780, respectively).

Declaration of generative AI use

During the preparation of this work, the author used Chat-GPT (OpenAI) solely to check grammar and polish wording. The author reviewed and edited all resulting text and takes full responsibility for the content of the published article.

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